

Electrochemistry

A Level Chemistry

Oxidation number

The **oxidation number** 氧化数 (also called the oxidation state) shows how many **electrons** 电子 an atom has lost or gained compared with the free element. You work it out using simple rules:

- an uncombined element has an oxidation number of 0.
- a simple ion has an oxidation number equal to its charge (so Mg^{2+} is +2).
- Group 1 is always +1, Group 2 is always +2.
- hydrogen is +1 (but -1 in metal hydrides).
- oxygen is -2 (but -1 in peroxides).
- fluorine is always -1.
- the oxidation numbers in a neutral compound add up to 0; in an ion they add up to the charge.

A Roman numeral shows the size of the oxidation number of an element, for example iron(II) means +2 and manganese(VII) in KMnO_4 means +7.

Redox in terms of electrons

A **redox** 氧化还原 reaction is one where electrons move from one species to another.

- **oxidation** 氧化 is the **loss** of electrons. The oxidation number goes **up**.
- **reduction** 还原 is the **gain** of electrons. The oxidation number goes **down**.

A useful memory aid is **OIL RIG**: Oxidation Is Loss, Reduction Is Gain.

Oxidation and reduction always happen together, because the electrons lost by one species are gained by another. This **electron transfer** 电子转移 is why we call it a redox reaction.

Using oxidation numbers to balance equations

Changes in oxidation number help you balance a redox equation:

1. find the element whose oxidation number **rises** (it is oxidised) and the one whose number **falls** (it is reduced).
2. the total rise must equal the total fall, because every electron lost is gained somewhere.
3. choose the ratio of the two species so the rise and fall match, then balance the rest of the equation.

Disproportionation

Disproportionation 歧化 is a special redox reaction in which the **same** element is both oxidised and reduced at the same time. For example, when chlorine reacts with cold water, some chlorine atoms are reduced (to Cl^-) and others are oxidised (to ClO^-).

Oxidising and reducing agents

- an **oxidising agent** 氧化剂 takes electrons away from another species. In doing so, it is itself reduced.
- a **reducing agent** 还原剂 gives electrons to another species. In doing so, it is itself oxidised.

So in any redox reaction, the oxidising agent causes the oxidation of the other species, while the reducing agent causes the reduction.